

**Generative Discovery and First-Principles Validation
of
Economical High-Entropy Carbide Electrocatalysts
for the Electrochemical CO₂-to-CO Reduction
Reaction**

Hyun-Ju Lee

Advisor: [Advisor Name]
[Department], [University]

June 2026

Abstract

The electrochemical reduction of carbon dioxide to carbon monoxide (CO₂RR-to-CO) is a key step toward closing the anthropogenic carbon cycle, but the most selective known catalysts (Au, Ag) are scarce and expensive. This thesis develops and applies an end-to-end *generative* discovery pipeline that searches directly for *economical* CO-evolving catalysts within the chemical space of high-entropy transition-metal carbides (HECs). A property-conditioned diffusion model (a fine-tuned MATTERGEN) is steered toward stable, metallic, CO-releasing and platinum-group-metal-free compositions; the generated pool is funneled through composition proxies, a machine-learned interatomic-potential (UMA) surface free-energy screen, and density-functional-theory (DFT) validation in QUANTUM ESPRESSO. Starting from 1024 generated structures the pipeline converges on six candidate HECs (Pr₃Y₂WC₇, La₅Y₄W₂C₉, Ce₂Y₂ZrWC₆, La₃Y₅(WC₃)₃, Y₂ZrWC₈, CeY₂ZrWC₇), each predicted to host a near-thermoneutral CO site ($|\Delta G_{\text{CO}^*}| \lesssim 0.02 \text{ eV}$) and CO-over-H selectivity. DFT confirms all six are metallic; projected densities of states resolve the conduction character as a carbon-2*p* network hybridized with the metal *d* sublattice. A complementary experimental validation campaign—materials synthesis, electrochemical CO₂RR testing, on-line gas analysis for CO/CO₂, ¹H NMR for liquid products, and (in-situ) FT-IR for adsorbed intermediates—is designed and presented; the corresponding results are reserved for completion. The work demonstrates that economic constraints can be embedded *at the generative stage*, shifting catalyst discovery from screening known materials toward designing affordable new ones.

Keywords: CO₂ reduction; high-entropy carbides; generative models; machine-learned interatomic potentials; density functional theory; non-precious electrocatalysts.

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Chapter 1

Introduction

1.1 Motivation: affordable CO₂-to-CO conversion

Electrochemical CO₂ reduction (CO₂RR) powered by renewable electricity offers a route to convert a waste greenhouse gas into chemical feedstocks. Among the accessible products, carbon monoxide is one of the most technologically valuable: it is the two-electron product with the highest demonstrated selectivity and is the entry point to Fischer–Tropsch and carbonylation chemistry. The selectivity of a metal electrode for CO is governed by how strongly it binds the key intermediates *COOH and *CO relative to adsorbed hydrogen *H; the classic survey of metal electrodes by Hori established that Au, Ag and Zn are the principal CO-evolving metals, whereas Cu uniquely proceeds to hydrocarbons and most other metals favor the competing hydrogen-evolution reaction (HER) [9]. The mechanistic basis of this behavior on copper, and the central role of the *COOH and *CO descriptors, is reviewed comprehensively by Nitopi *et al.* [8].

The difficulty is economic rather than purely chemical: the most CO-selective metals (Au, Ag) are scarce and costly, which is incompatible with the terawatt scale at which CO₂ conversion would need to operate to matter for climate. There is therefore a pressing need for catalysts that reproduce the favorable adsorption-energy landscape of the coinage metals using earth-abundant elements.

1.2 Why high-entropy carbides

Two materials-design ideas are combined in this thesis. First, transition-metal carbides (TMCs) have long been known to display “platinum-like” surface chemistry: the seminal observation of Levy and Boudart that tungsten carbide mimics platinum in several catalytic reactions [12] initiated decades of work on TMCs as inexpensive substitutes for precious metals, and molybdenum carbide in particular is an established, selective catalyst for reducing CO₂ to CO [13]. Carbon donates electron density into the metal *d* band and reshapes the surface density of states, tuning adsorption strengths into useful windows.

Second, *high-entropy* materials—near-equimolar solid solutions of four or more elements—provide a near-continuous distribution of local adsorption environments. Batchelor *et al.* argued that this continuum is precisely what an electrocatalyst needs, because a small fraction of optimally-binding sites can dominate the measured activity [10]; high-entropy alloys have since been realized experimentally as CO₂/CO reduction catalysts [11]. High-entropy carbides are synthesizable single-phase materials in their own right, as demonstrated by the entropy-descriptor discovery of refractory five-metal carbides [14]. Combining the carbide “platinum-like” electronic structure with high-entropy site diversity—while restricting the palette to cheap metals—defines the search space of this work.

1.3 Why a generative pipeline

Conventional computational catalyst discovery screens *existing* databases. The combinatorial space of multi-metal carbides is far too large to enumerate, and the affordable subset is sparse within it. We instead adopt a *generative* strategy: a diffusion model that has learned the distribution of stable inorganic crystals [1] is fine-tuned so that its sampling is *steered* toward the target property values—including an explicit *economic* constraint that penalizes platinum-group and heavy-rare-earth content. Affordability thus becomes a generation objective rather than a post-hoc filter. Because the model proposes novel structures, the expensive part of discovery (DFT) is reserved for a handful of already-promising candidates, reached through a funnel of increasingly accurate but increasingly costly evaluators.

1.4 Objectives and structure

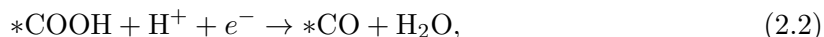
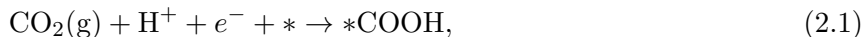
The objectives of this thesis are: (i) to construct an economic-constrained generative pipeline for CO₂RR-to-CO HEC catalysts; (ii) to validate the top candidates with first-principles DFT for stability and metallic conductivity, resolving the orbital character of the conduction states; and (iii) to design the experimental campaign that will confirm the computational predictions. Chapter 2 reviews the theoretical background; Chapter 3 details the computational methodology; Chapter 4 presents the computational results; Chapter 5 sets out the planned experimental validation; Chapter 6 discusses the findings and limitations; and Chapter 7 concludes.

Chapter 2

Theoretical Background

2.1 Thermodynamics of CO₂-to-CO and the computational hydrogen electrode

The two-electron pathway to CO proceeds through a single carboxyl intermediate,



Reaction free energies of the electrochemical steps are evaluated with the computational hydrogen electrode (CHE) of Nørskov *et al.* [7], in which the chemical potential of a proton–electron pair is referenced to gas-phase H₂ at 0 V versus the reversible hydrogen electrode:

$$\mu(\text{H}^+) + \mu(e^-) = \frac{1}{2} \mu(\text{H}_2) - eU. \quad (2.4)$$

The adsorption free energies used throughout are

$$\Delta G_{\text{CO}*} = E(*\text{CO}) - E(*) - E(\text{CO}_{(\text{g})}) + \Delta_{\text{CO}}, \quad (2.5)$$

$$\Delta G_{\text{COOH}*} = E(*\text{COOH}) - E(*) - E(\text{CO}_{2(\text{g})}) - \frac{1}{2} E(\text{H}_{2(\text{g})}) + \Delta_{\text{COOH}}, \quad (2.6)$$

$$\Delta G_{\text{H}*} = E(*\text{H}) - E(*) - \frac{1}{2} E(\text{H}_{2(\text{g})}) + \Delta_{\text{H}}, \quad (2.7)$$

where the Δ terms are standard zero-point and entropic corrections. A good CO-evolving site requires $\Delta G_{\text{CO}*}$ near zero (CO is released as product rather than poisoning the surface), $\Delta G_{\text{COOH}*}$ low enough to activate CO₂, and—crucially for selectivity— $\Delta G_{\text{COOH}*} < \Delta G_{\text{H}*}$, so that CO₂RR out-competes HER.

2.2 Descriptors, the Sabatier principle and carbides

Because adsorption energies on related surfaces scale with one another, a single descriptor (here the best-site $\Delta G_{\text{CO}*}$) captures most of the activity trend, and the optimum is a Sabatier maximum at intermediate binding. In carbides, carbon p states hybridize with metal d states and shift the d -band, which is the microscopic origin of the “platinum-like” behavior of WC [12] and of the CO selectivity of Mo₂C [13]. A metallic density of states at the Fermi level is a prerequisite for an electrocatalyst, so verifying metallicity by DFT is an essential validation step.

2.3 Generative diffusion models for crystals

MATTERGEN [1] is a diffusion model that jointly denoises the atomic coordinates, atom types and lattice of a crystal, trained on a large database of stable inorganic compounds. Conditioning

is achieved by fine-tuning property-embedding adapters so that classifier-free guidance biases sampling toward target values of one or more properties; the guidance strength w trades diversity against on-target yield. This is the mechanism we exploit to embed both catalytic and *economic* targets directly into generation.

2.4 Machine-learned interatomic potentials

Evaluating surface free energies for thousands of candidates by DFT is intractable. We use a universal machine-learned interatomic potential (MLIP), the Universal Model for Atoms (UMA) [2], whose catalysis behavior derives from training on the Open Catalyst 2020 dataset [3]. The MLIP provides near-DFT adsorption energetics at a tiny fraction of the cost, serving as the intermediate rung between cheap composition proxies and full DFT.

2.5 Density functional theory and pseudopotentials

Final validation uses Kohn–Sham DFT with the PBE exchange–correlation functional [6] as implemented in the plane-wave code QUANTUM ESPRESSO [4], with pseudopotentials from the Standard Solid-State Pseudopotential (SSSP) library [5]. Configurational disorder of the high-entropy sublattice is represented, where required, by special quasirandom structures (SQS) [15].

Chapter 3

Computational Methodology

The discovery pipeline is a five-stage funnel of increasing fidelity and cost (Fig. 4.1a). Each stage is described below.

3.1 Stage A — economic-conditioned generation

A base MATTERGEN checkpoint was fine-tuned with nine conditioning properties: stability (energy above hull), metallicity (driving the band gap to zero), the primary CO-release descriptors (best-site CO affinity and the fraction of CO-selective sites), CO₂ activation (mean *COOH affinity), CO-vs-H selectivity, high mixing entropy, a metallic-packing proxy, and—decisively—the *expensive-metal fraction*, defined as the fraction of metal atoms that are platinum-group (Ru, Rh, Pd, Os, Ir, Pt) or heavy rare-earth, which is driven to zero. Training labels were computed for every transition-metal carbide in the `alex_mp_20` dataset by literature-informed composition proxies. Fine-tuning ran for 600 epochs; the validation loss is shown in Fig. 4.2. Conditional generation was performed at two guidance strengths ($w = 1.0$ and $w = 2.5$), yielding 1024 candidate structures.

3.2 Stage B — cheap composition screen

Every generated structure was re-evaluated with the composition proxies and filtered to retain those that are (i) genuine transition-metal carbides, (ii) within the CO-release window, (iii) high-entropy ($n_{\text{metal}} \geq 3$, mixing entropy $\geq 1.0 k_{\text{B}}$), and (iv) economically clean (zero expensive-metal fraction). This narrowed the pool to 104 survivors (90 distinct compositions).

3.3 Stage C — MLIP surface free-energy screen

For the 24 highest-ranked distinct compositions, low-index surface terminations were enumerated (maximum Miller index 1); the most stable termination per facet was relaxed with the bottom half fixed, each adsorbate (*CO, *COOH, *H) was placed at every symmetry-distinct site, and the adslabs were relaxed with the UMA potential [2]. Free energies followed the CHE expressions of Section 2. Gas-phase and clean-slab references were computed consistently with the total-energy head of the same potential so that the slab contribution cancels; non-physical, non-converged sites ($|\Delta G| > 10 \text{ eV}$) were discarded. Each candidate’s descriptors aggregate over *all* computed facets and sites, and the candidates were ranked by proximity of their best site to the CO optimum and by CO-selective site density.

3.4 Stage D — DFT validation and electronic structure

The six top-ranked candidates were validated with QUANTUM ESPRESSO [4]: variable-cell relaxation, followed by a self-consistent calculation on the relaxed geometry, and a density-of-states (DOS) calculation. Plane-wave cutoffs of 60 Ry (wavefunctions) and 480 Ry (charge density) were used with Marzari–Vanderbilt cold smearing (0.02 Ry) and Monkhorst–Pack k -meshes of density $\sim 30 \text{ \AA}/a$. SSSP pseudopotentials [5] were used for C, Y, Zr, W; for the rare earths (La, Ce, Pr) norm-conserving pseudopotentials with the $4f$ shell frozen in the core (trivalent) were used, which is physically appropriate for the localized $4f$ states of these metallic carbides. Element-projected DOS was obtained by projecting the converged Kohn–Sham states onto atomic orbitals (no additional self-consistency), summed per element to quantify each sublattice’s contribution to the states at the Fermi level.

3.5 Stage A' — solid-solution supercells

To represent explicit chemical disorder, special quasirandom structures [15] of the high-entropy shortlist were built on rock-salt (MC) or hexagonal (M_2C) frameworks by Monte-Carlo matching of the metal-sublattice pair-correlation functions; these supercells are the substrate for future explicit-disorder surface studies.

3.6 Software

Structure manipulation used `pymatgen` [16] and the Atomic Simulation Environment [17]; the MLIP screen used the `fairchem`/UMA implementation [2].

Chapter 4

Computational Results

4.1 Generation and the discovery funnel

Fine-tuning converged smoothly (Fig. 4.2), and conditional generation produced 1024 structures of which 1004 are transition-metal carbides. Their composition-descriptor distributions (Fig. 4.3) show that the economic conditioning was effective: the expensive-metal fraction is essentially zero across the pool, and the metal palette is dominated by inexpensive early transition metals and light rare earths (W, Y, Zr, Ce, La, Pr). The cheap screen retained 104 economical CO-releasing high-entropy carbides. The full funnel ($1024 \rightarrow 1004 \rightarrow 90 \rightarrow 24 \rightarrow 6$) is summarized in Fig. 4.1a.

4.2 MLIP surface screen

The UMA surface screen evaluated all low-index facets of the 24 candidates (Fig. 4.4). The best-site ΔG_{CO^*} of the leading candidates sits within a few tens of meV of the CO optimum, and the best site is found on a variety of facets rather than a single preferred plane—consistent with the high-entropy design intent of distributing favorable sites. The six top-ranked candidates, all CO-over-H selective at their best site, are listed in Table 4.1.

4.3 DFT validation: stability and metallicity

All six candidates relaxed to low residual pressure ($|P| < 0.5 \text{ kbar}$) and are *metallic*, with a finite density of states at the Fermi level ranging from 1.7–12.4 states/eV per cell (Fig. 4.6, Table 4.1). Metallic conductivity, a prerequisite for electrocatalysis, is therefore confirmed by first principles.

4.4 Conduction character from projected DOS

The element-projected DOS (Fig. 4.5) resolves *which* sublattice carries the Fermi-level states. Two regimes emerge: carbon-rich candidates ($\text{Pr}_3\text{Y}_2\text{WC}_7$, $\text{Ce}_2\text{Y}_2\text{ZrWC}_6$) are dominated by carbon-*2p* character at E_F ($\sim 60\%$), describing a conductive carbide network, whereas the tungsten-rich $\text{La}_3\text{Y}_5(\text{WC}_3)_3$ and the rare-earth-free Y_2ZrWC_8 are metal-*d* dominated (Y/La/W). The per-element shares are plotted in Fig. 4.1d. This hybridized C-*p*/metal-*d* conduction is exactly the electronic signature expected for “platinum-like” carbide catalysts [12].

4.5 Summary table

Table 4.1 collects the surface activity descriptors and the DFT validation for the six final candidates; the corresponding composite figure is Fig. 4.1.

Table 4.1: Master summary of the six final economical high-entropy carbide candidates: surface CO-release descriptor (ΔG_{CO^*} of the best site, from the UMA screen), CO-over-H selectivity, and DFT-validated Fermi energy, density of states at E_{F} , dominant conduction element, and residual pressure. All six are metallic.

Candidate	ΔG_{CO^*} (eV)	CO>H	E_{F} (eV)	DOS(E_{F})	dominant	metallic	P (kbar)
$\text{Pr}_3\text{Y}_2\text{WC}_7$	−0.001	yes	12.00	7.4	C	yes	+0.15
$\text{La}_5\text{Y}_4\text{W}_2\text{C}_9$	−0.001	yes	11.52	9.8	C	yes	−0.10
$\text{Ce}_2\text{Y}_2\text{ZrWC}_6$	+0.007	yes	12.72	6.7	C	yes	−0.29
$\text{La}_3\text{Y}_5(\text{WC}_3)_3$	−0.008	yes	12.24	12.4	Y	yes	−0.32
Y_2ZrWC_8	−0.010	yes	10.52	1.7	W	yes	+0.41
$\text{CeY}_2\text{ZrWC}_7$	−0.011	yes	12.85	3.3	C	yes	−0.13

4.6 Solid-solution supercells and outstanding computational work

Twelve special quasirandom structures of the high-entropy shortlist were constructed (Stage A', Sec. 3) as explicit-disorder supercells on rock-salt and hexagonal frameworks, each annealed to a low residual metal-sublattice pair-correlation error. These are the substrate for the explicit-disorder surface studies below. The following first-principles results are *in progress* and will be incorporated upon completion:

- [DFT (QE) surface free energies ΔG_{CO^*} , ΔG_{COOH^*} , ΔG_{H^*} on the relaxed facets / SQS supercells, confirming the MLIP-level activity ranking and the $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$ selectivity at the DFT level – to be added when the calculation completes]
- [DFT formation / decomposition energies and convex-hull stability of the six candidates, beyond the residual-stress check reported in Table 4.1 – to be added when the calculation completes]
- [the full computational-hydrogen-electrode free-energy diagram and limiting potential U_L of the best site of each candidate – to be added when the calculation completes]
- [d -band-center / projected-crystal-orbital analysis linking the conduction character (Fig. 4.5) to the adsorption descriptors – to be added when the calculation completes]

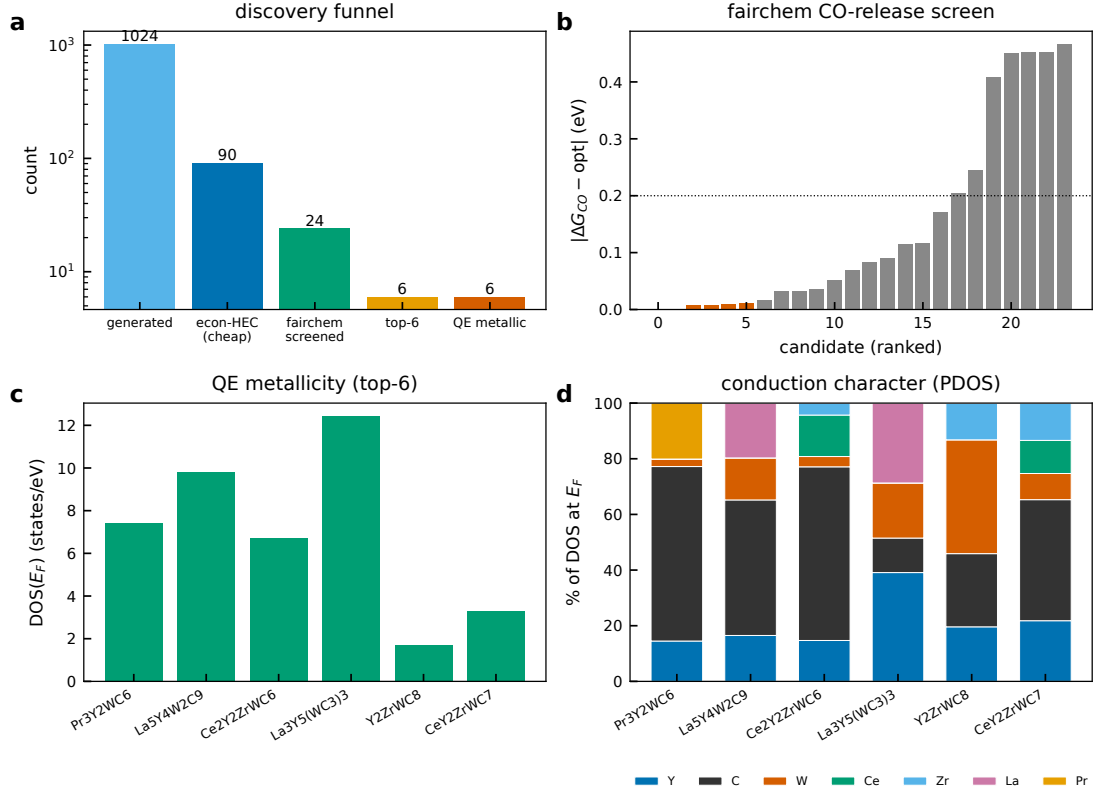


Figure 4.1: End-to-end discovery synthesis. (a) The discovery funnel from 1024 generated structures to six DFT-validated metallic candidates. (b) UMA surface CO-release descriptor of the screened candidates (top-6 highlighted). (c) DFT density of states at the Fermi level for the top-6. (d) Element-projected conduction character (per-element share of DOS at E_F).

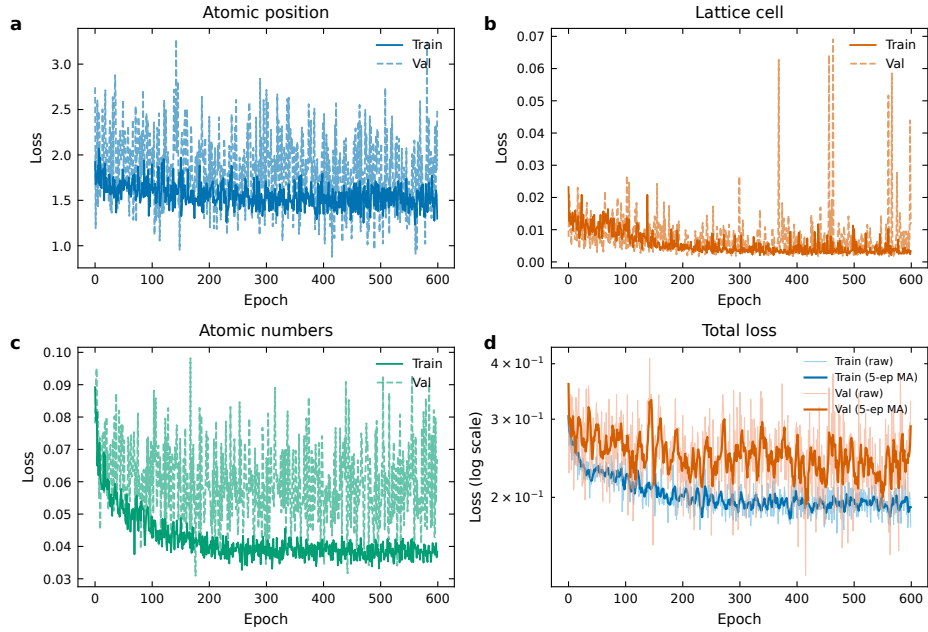


Figure 4.2: Fine-tuning of the economic-conditioned generative model: training/validation loss and per-component (position, lattice, atom-type) losses over 600 epochs.

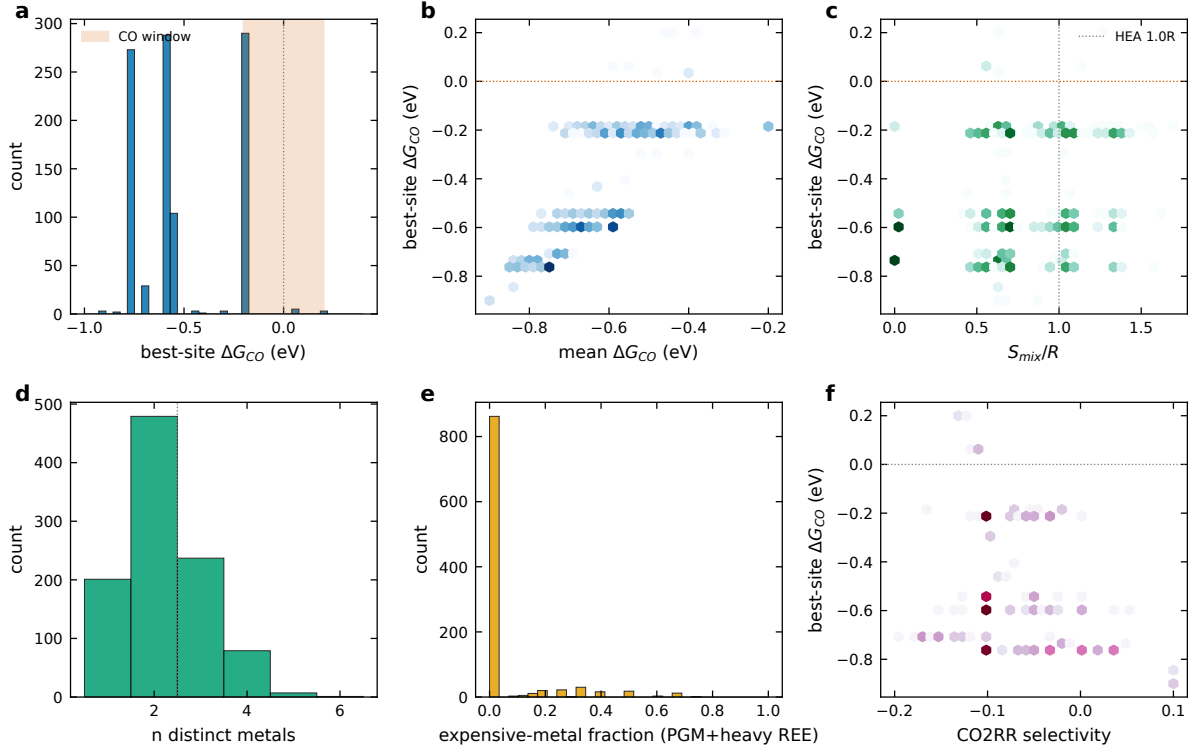


Figure 4.3: Composition statistics of the generated carbide pool: (a) best-site CO affinity with the CO-release window; (b,c) descriptor correlations; (d) metal-count distribution; (e) expensive-metal fraction (economic constraint); (f) CO₂RR selectivity.

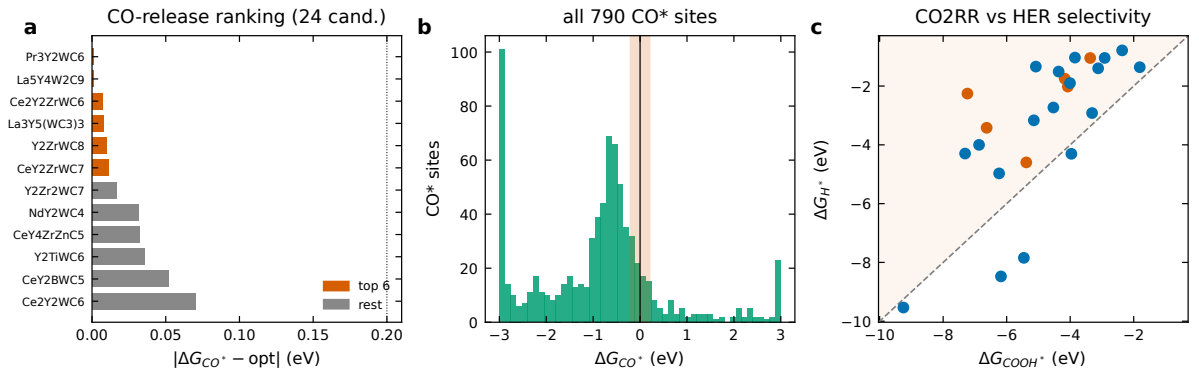


Figure 4.4: UMA surface screen: (a) candidates ranked by proximity of the best site to the CO optimum; (b) pooled ΔG_{CO^*} distribution with the CO-release window; (c) CO-vs-H₂ selectivity (ΔG_{COOH^*} versus ΔG_{H^*}).

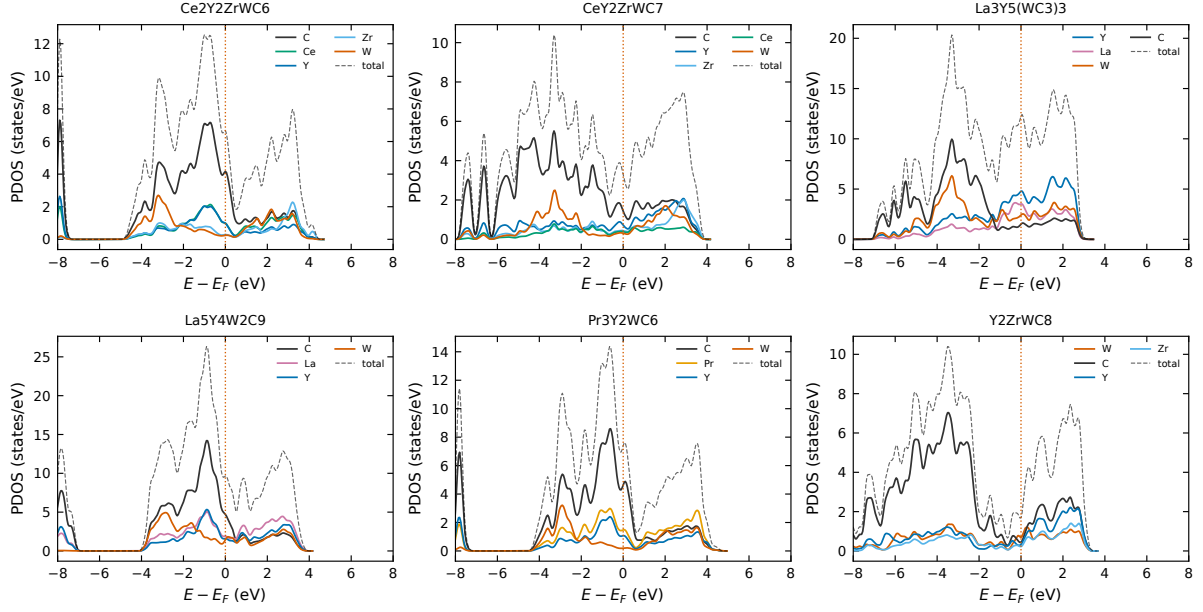


Figure 4.5: Element-projected density of states (PDOS) of the six DFT-validated candidates versus $E - E_F$. Finite states at E_F (dashed line) confirm metallicity; the elemental decomposition identifies the conduction character.

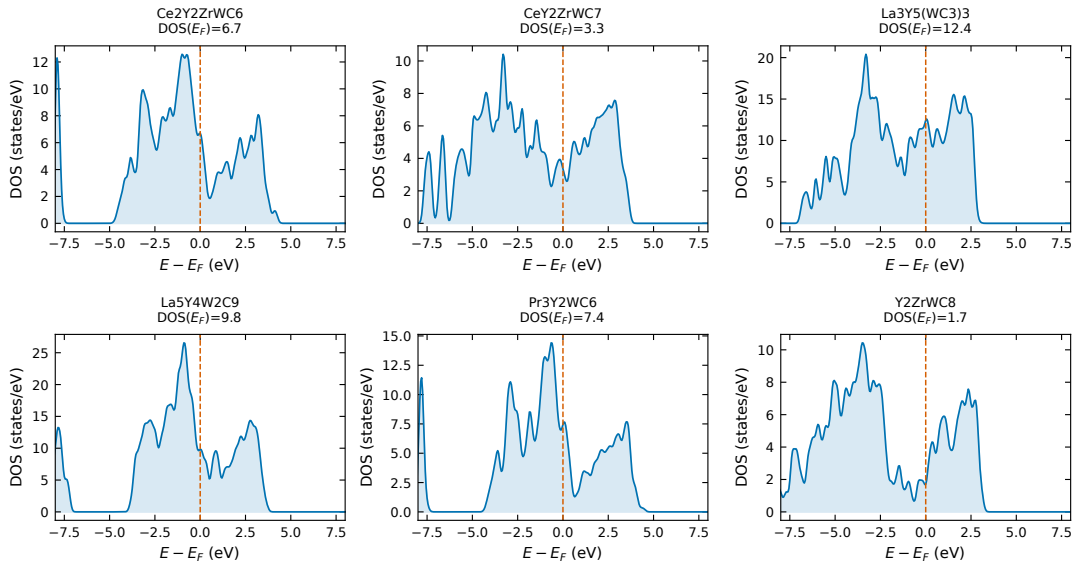


Figure 4.6: Total density of states of the six candidates near the Fermi level (DFT, PBE).

Chapter 5

Planned Experimental Validation

The computational pipeline yields concrete, falsifiable predictions: the six candidates should (i) be synthesizable as single-phase metallic carbides and (ii) electroreduce CO_2 to CO with selectivity exceeding HER. This chapter specifies the experimental program designed to test them. Results are reserved until the campaign is complete.

5.1 Materials synthesis

The shortlisted high-entropy carbides will be synthesized by high-temperature carburization of ball-milled metal/oxide precursors (or by arc-melting followed by carburization), targeting the SQS-predicted near-equimolar metal sublattice. Phase purity and the single-phase rock-salt/hexagonal structure will be verified by powder X-ray diffraction (XRD); homogeneity of the metal distribution by SEM/EDS and STEM-EDS mapping; and surface composition/oxidation state by X-ray photoelectron spectroscopy (XPS). **[Synthesis and structural characterization – to be updated upon experimental completion]**

5.2 Electrochemical CO_2 RR testing

Catalyst inks will be drop-cast onto gas-diffusion or glassy-carbon electrodes and tested in a gas-tight H-cell (and an MEA/flow cell for higher current density) in CO_2 -saturated aqueous bicarbonate electrolyte. Linear-sweep voltammetry and chronoamperometry at a series of applied potentials will quantify the partial current densities; the geometric and electrochemically-active-surface-area-normalized activities will be reported. **[Polarization curves and Faradaic efficiency vs. potential – to be updated upon experimental completion]**

5.3 Product analysis

- **Gas products (CO , H_2 , residual CO_2).** The cell headspace/outlet will be sampled on-line into a gas chromatograph (TCD + FID with methanizer) to quantify CO and H_2 and thus the CO Faradaic efficiency and the CO/ H_2 selectivity; CO_2 conversion will be tracked from the inlet/outlet balance. On-line differential electrochemical mass spectrometry (DEMS) will provide potential-resolved CO/ H_2 onset. **[GC/DEMS gas-product quantification – to be updated upon experimental completion]**
- **Liquid products.** The catholyte will be analyzed by quantitative ^1H nuclear magnetic resonance (NMR) spectroscopy with water suppression and an internal standard (e.g. DMSO/DSS) to detect and quantify any formate or minor liquid products, confirming the expected dominance of the two-electron CO pathway. **[^1H NMR liquid-product quantification – to be updated upon experimental completion]**

Table 5.1: Planned comparison of computational predictions with experimental observables.

Predicted (this work)	Experimental probe	Status
Single-phase HEC, equimolar metals	XRD, STEM-EDS	[result – to be updated upon experimental completion]
Metallic (finite DOS at E_F)	4-probe / EIS	[result – to be updated upon experimental completion]
CO-selective, $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$	GC, ^1H NMR, FE	[result – to be updated upon experimental completion]
*COOH→*CO pathway	in-situ FT-IR	[result – to be updated upon experimental completion]
CO ₂ conversion	inlet/outlet GC balance	[result – to be updated upon experimental completion]

- **Adsorbed intermediates.** In-situ attenuated-total-reflection surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS / FT-IR) will be used to detect the *COOH and linearly/bridge-bound *CO bands as a function of potential, providing a direct mechanistic link to the computed ΔG_{COOH^*} and ΔG_{CO^*} descriptors. **[In-situ FT-IR of *COOH/*CO intermediates – to be updated upon experimental completion]**

5.4 Stability and post-mortem analysis

Extended electrolysis will assess durability; XRD, XPS and microscopy before/after testing will probe surface oxidation, leaching of the rare-earth components, and carbide-network integrity. **[Durability and post-mortem characterization – to be updated upon experimental completion]**

5.5 Validation logic

The experiments are designed to test the computational predictions point-by-point: metallic conductivity (four-point probe / electrochemical impedance) against the DFT DOS; CO selectivity (GC/NMR) against the computed ΔG_{CO^*} and $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$ selectivity criterion; and the mechanistic intermediates (FT-IR) against the descriptor ladder. A predicted-vs-measured comparison table is reserved for completion (Table 5.1).

Chapter 6

Discussion

6.1 Why this pipeline, and what it buys

The central methodological claim of this thesis is that *economic constraints belong at the generative stage*. By fine-tuning the diffusion model on an expensive-metal-fraction label and guiding it to zero, the affordable subspace is sampled *by construction*, rather than being discovered by filtering an unconstrained pool in which it is sparse. The resulting funnel spends DFT only on six structures while still surveying $\sim 10^3$ candidates, with each rung (composition proxy $\rightarrow$ MLIP $\rightarrow$ DFT) adding accuracy where the previous rung has already removed the bulk of the search space. This is a generalizable template: swapping the conditioning descriptors retargets the same machinery to a different reaction.

6.2 Interpreting the candidates

All six candidates combine cheap early transition metals (Y, Zr, W) with light rare earths (La, Ce, Pr) on the metal sublattice of a carbide. The DFT-resolved conduction character (Fig. 4.5) shows a continuum from carbon-2*p*-network metals to metal-*d* metals, mirroring the “platinum-like” picture in which carbon reshapes the metal *d* states [12]. The near-thermoneutral ΔG_{CO^*} and the $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$ selectivity place these materials in the same descriptor window that makes Au/Ag/Zn CO-selective [9], but with earth-abundant constituents—which is precisely the design goal. The success of high-entropy alloys as experimental CO₂/CO reduction catalysts [11, 10] and of Mo₂C for selective CO₂-to-CO [13] lends external plausibility to the carbide+HEC route.

6.3 Limitations and honest caveats

Several caveats must be stated. (i) The light rare earths (La, Ce, Pr, Y) are economically “inexpensive” by the chosen descriptor, but rare-earth carbides are unusual CO₂RR catalysts and may be oxophilic and air/moisture sensitive; experimental robustness is an open question that the planned stability tests directly address. (ii) The MLIP surface screen, although calibrated to cancel reference errors, evaluates rare-earth-carbide surfaces that lie outside the bulk of its training distribution; the strongly-bound *COOH energies in particular should be read qualitatively, and the activity ranking leans on the more robust ΔG_{CO^*} descriptor. (iii) The composition proxies used in Stages A–B are literature-informed estimates intended for coarse filtering, not quantitative prediction. (iv) The DFT treats the rare-earth 4*f* electrons as frozen trivalent cores; while appropriate for these metallic carbides, 4*f*-in-valence or DFT+*U* treatments would be needed for any property sensitive to *f*-states. (v) The present DFT validates *bulk* stability and metallicity; explicit-disorder *surface* free energies in DFT (on the SQS supercells of Stage A') are the natural next computational step, beyond the MLIP-level surface screen reported here.

These limitations motivate, rather than undermine, the experimental campaign of Chapter 5, which is the arbiter of the predictions.

6.4 Outlook

Beyond confirming the six candidates, the pipeline supports an active-learning loop in which experimental and higher-level DFT data recalibrate the composition proxies and the MLIP, tightening successive generations. The same economic-conditioned generator could be retargeted to CO₂-to-formate or to multi-carbon products by changing the conditioning descriptors.

Chapter 7

Conclusion

This thesis built and applied an end-to-end generative pipeline that designs *affordable* high-entropy carbide catalysts for the CO₂-to-CO reaction by embedding an economic constraint directly into a property-conditioned diffusion model and validating the results through a funnel of a machine-learned potential and first-principles DFT. From 1024 generated structures, six platinum-group-metal-free high-entropy carbides emerged, each predicted to host a near-thermoneutral CO site with CO-over-hydrogen selectivity, and all confirmed by DFT to be metallic with a hybridized carbon-*p*/metal-*d* conduction character. A complete experimental validation program—synthesis, electrochemical CO₂RR, on-line gas analysis, ¹H NMR, and in-situ FT-IR—has been designed to test these predictions, and its results will complete the work. The broader contribution is a reusable demonstration that affordability can be a *generative* design objective for catalyst discovery.

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